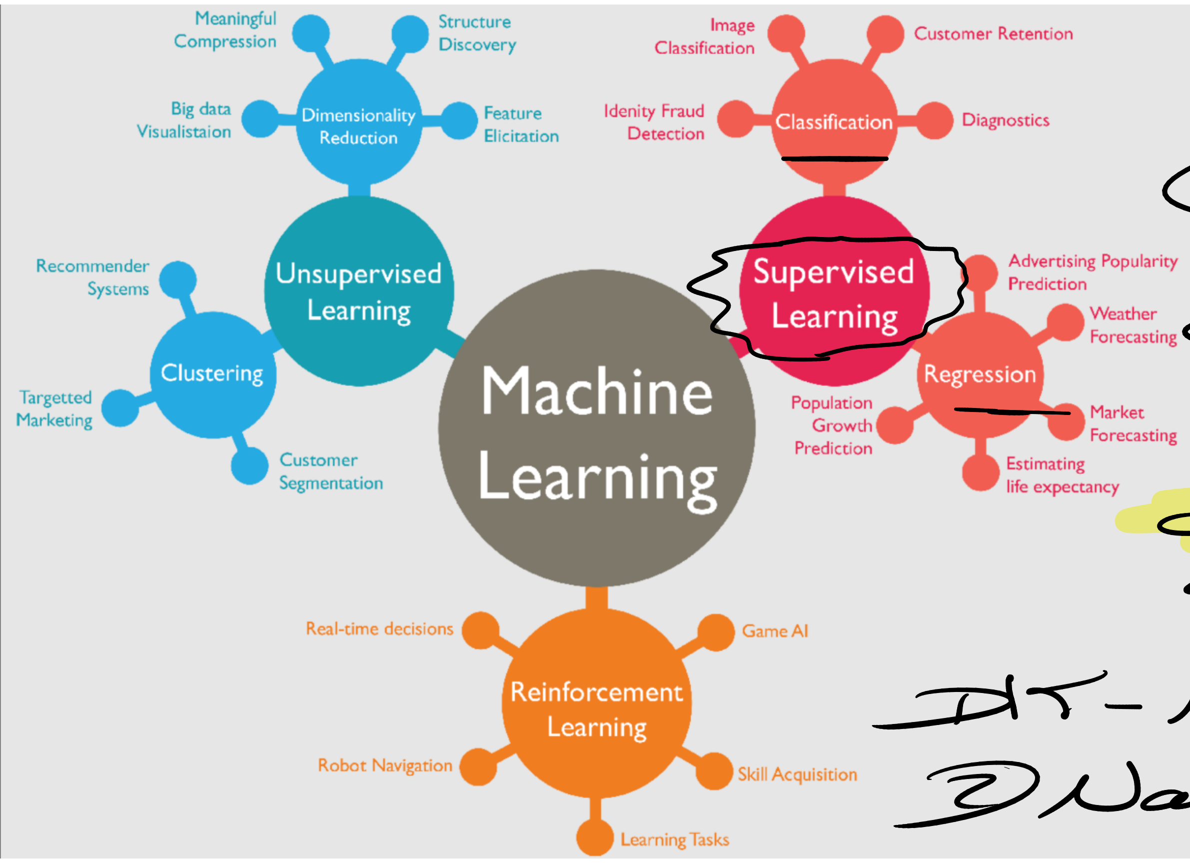


x, y

Cont.
→ Reg

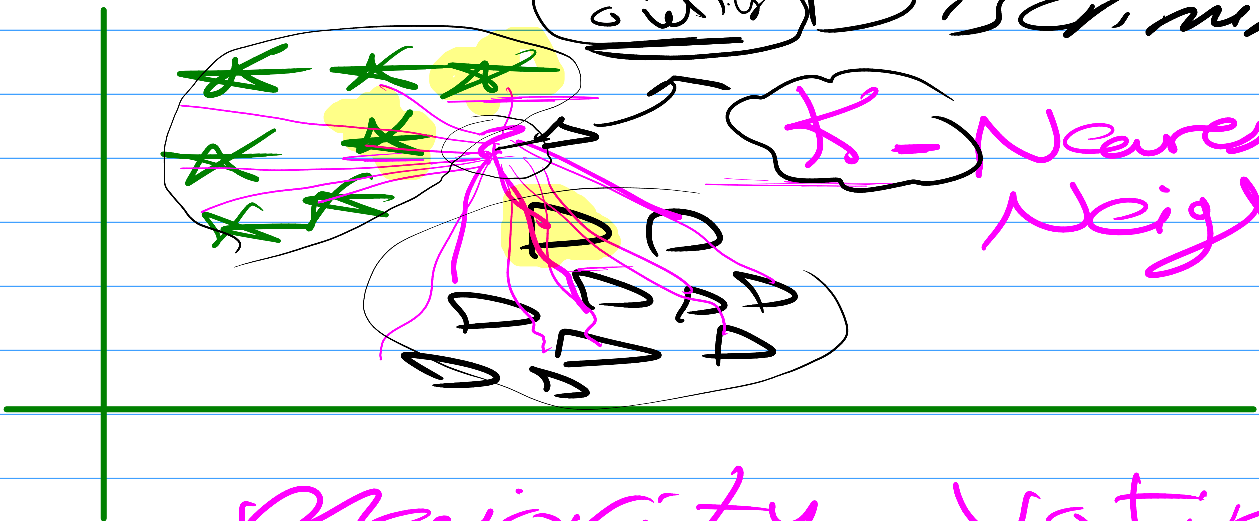
discrete
→ Classifiat

DT - NN

Naive Bayes

outlier Discrimination

K-Nearest Neighbor

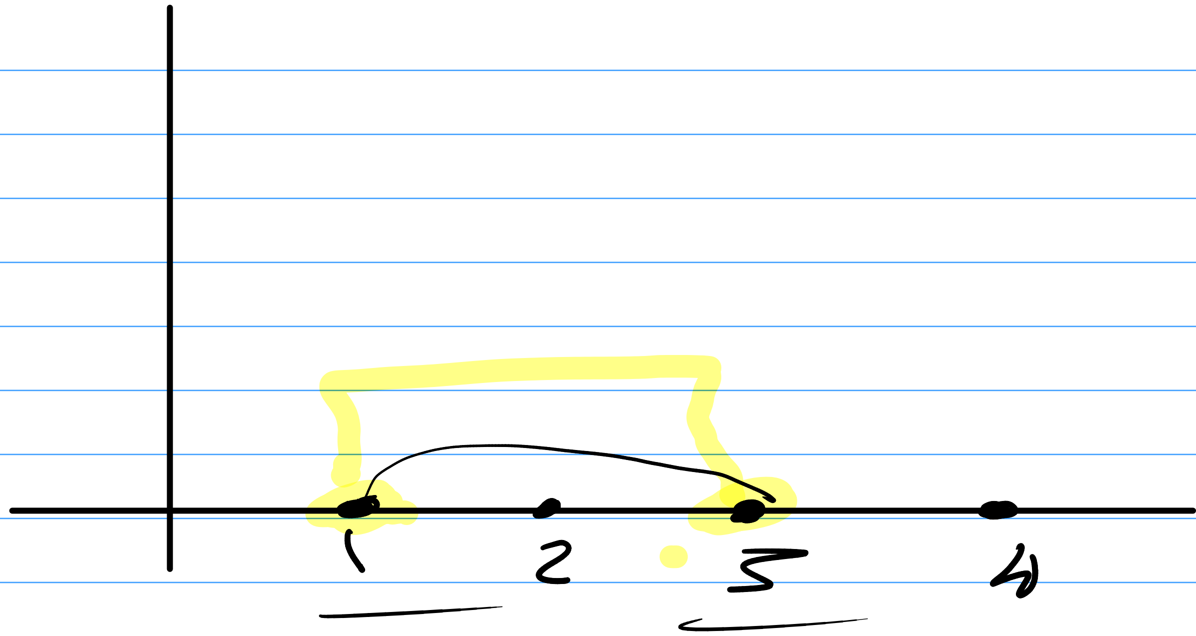


Majority Voting
→ (A)

(A) How to measure

Distance
→ distance↓

similarity↑



Distance = 2

① ? — *

① Manhattan distance

$$\sum_{i=1} \left| X_{??} - X_{iD} \right|$$

5 4

② Square Euclidean Distance

5 → 25

$$\left(X_{??} - X_{iD} \right)^2$$

2 4

→ الفرق يزداد اكثر

الصغير يزداد صغر

③ Euclidean

$$\sqrt{(x_2 - x_1)^2}$$

$$x_2 - x_1$$

??

④ Minkowski distance

Generalization

$$\sum_{i=1}^{n} \left(|x - y|^p\right)^{\frac{1}{p}}$$

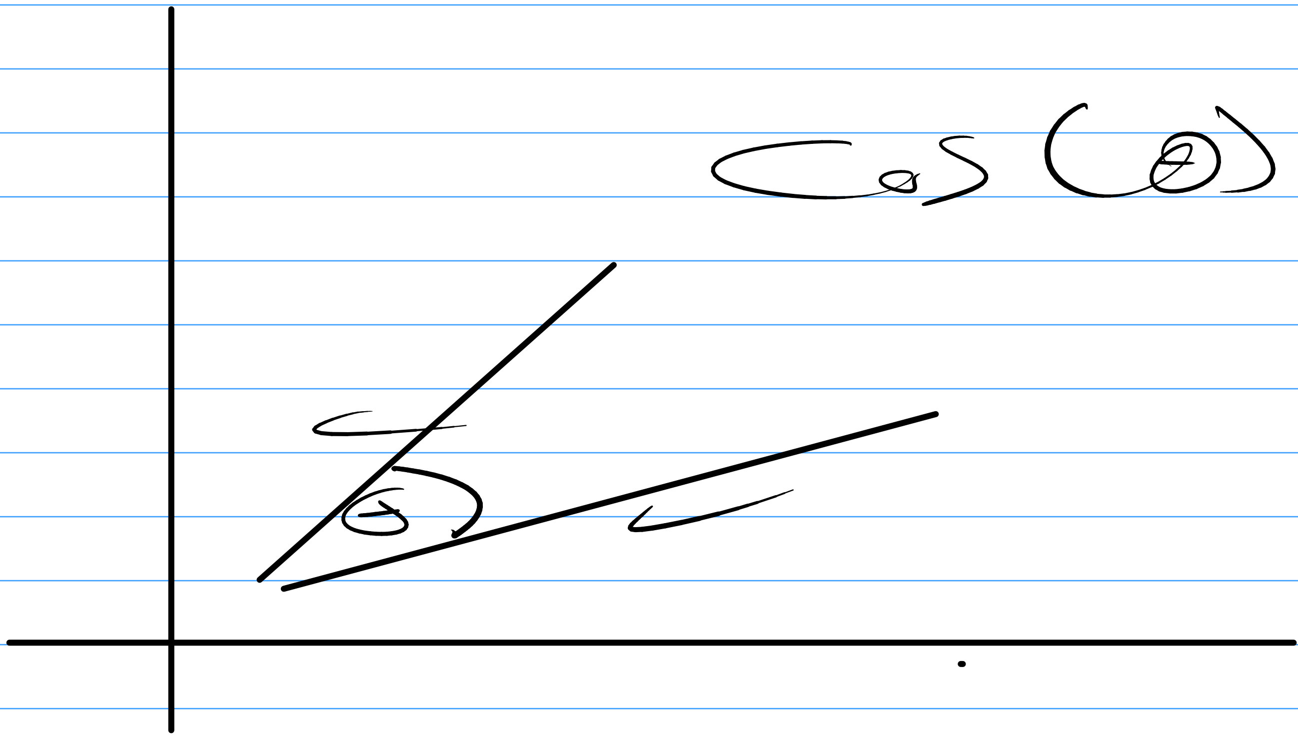
⑤ Similarity

x_1	x_2	x_3	x_4	y
3	2	1	5	0
2	2	1	2	1
2	1	5	5	1

$x_1 \quad x_2 \quad x_3 \quad x_4 \rightarrow Y$??

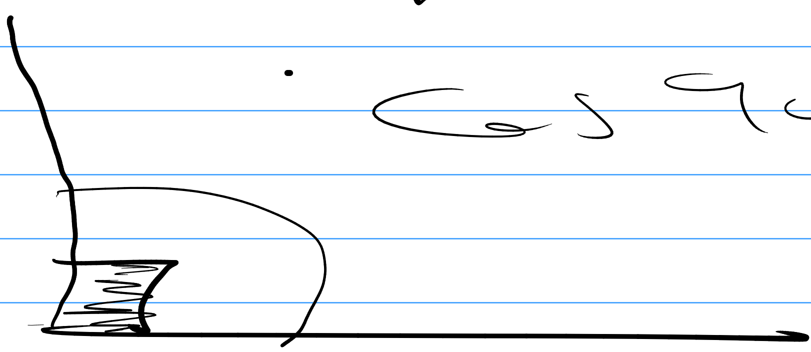
$\boxed{3 \quad 2 \quad 2 \quad 5}$

$[3 \quad 2 \quad 2 \quad 5]$



$\cos(0) = 1$

$\cos(90) = 0$



$\cos 90 = 0$

$\cos(0) = +1$

$\cos(90)$

$\cos(180)$

$\cos(0) = 1$

$x \qquad x_{22}$

$A \qquad B$

$$\cos(\theta) = \frac{A \cdot B}{\|A\| \, \|B\|}$$

$$\sqrt{(\ )^2 (\ )^2 (\ )^2}$$

$\cos(\theta)$

0

1

KNN

non-Parametric algorithms

$K = ???$ → 3, 4, 5

$$\underline{y} = \overset{\beta_1,\ w_1}{\underline{a}}\,\underline{X} + \overset{\beta_0,\ w_0}{\underline{b}}$$

slope — intercept

Majority Voting

Binary classification task — even, odd

$K =$ any number / even / odd

class

2 1 + 0

Multi class
Classification

K = 4 Classes

S → 2 1
2 2
1 3

K = 3 K = 6
K = 4
K = 5

trial & error

Accuracy ↑ → K

cross validation
gridsearch
K-Folds

Similarity $\neq$ distance

$\cos(\theta) = 1$

distance $\cos(\theta)$

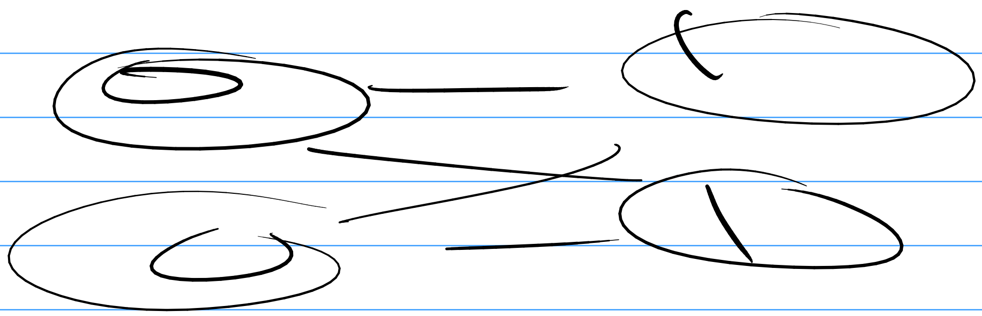
$= 1 - \text{Similarity}$

distance $= 1 - 1$

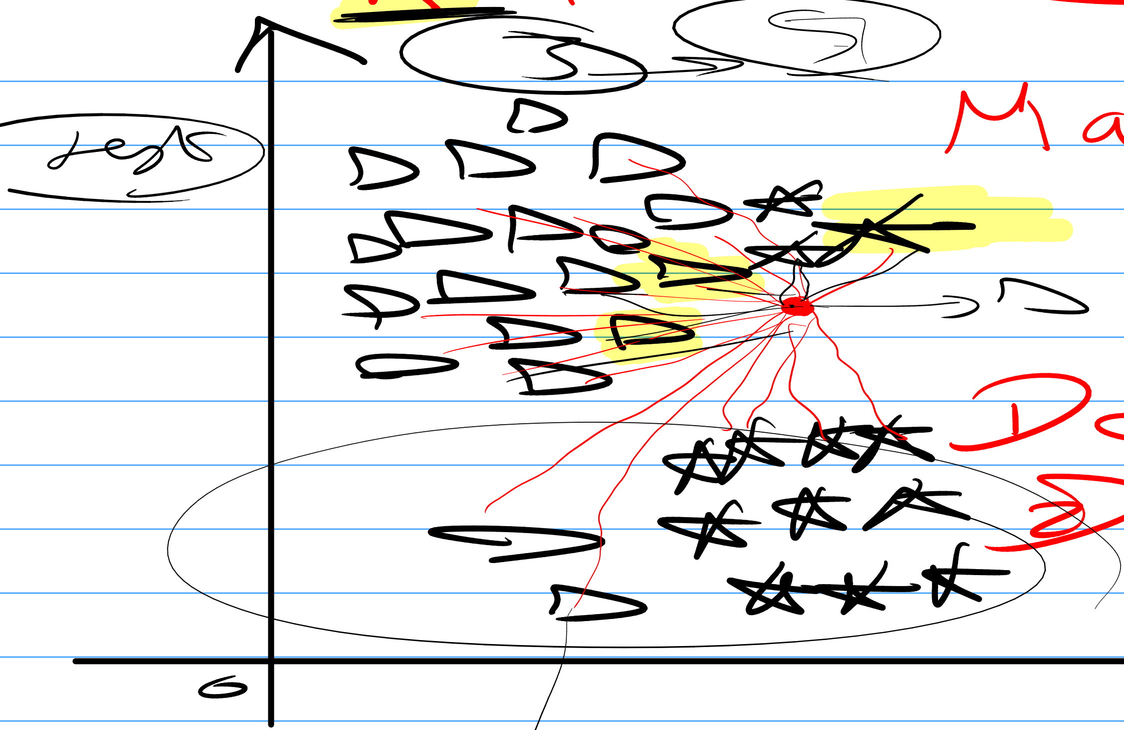
$=$ zero

distance $= 1 - 0$

$= 1$



K-NN   K=3

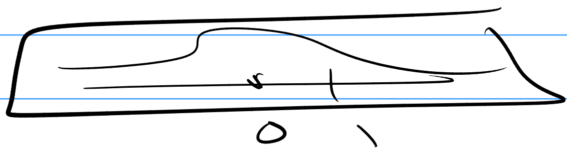


Majority Voting
D

Distance ≥ Majority

uniform K_NN

standardization

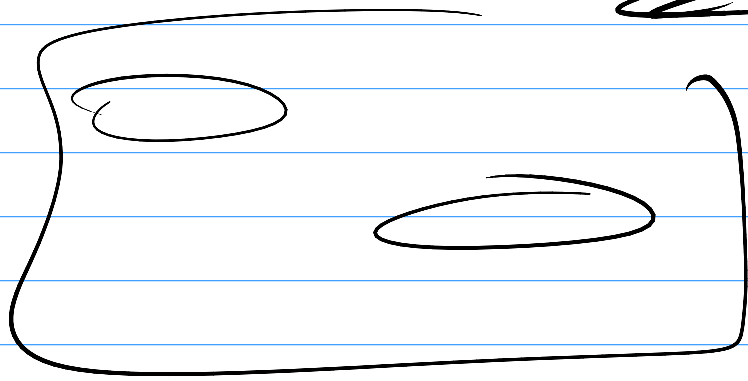


Logistic Reg.

model.predict(X)

$\sigma(aX+b)$

model.predict(X)



Iris dataset

builtin

↳ educational purpose

Projects

How to KNN

project

EPA - P —— iris

breast_cancer()

↳ 30 feature

B
M

عكس الحبث

30

Tuning

area

Convex

K Folds → 5

Imbalance

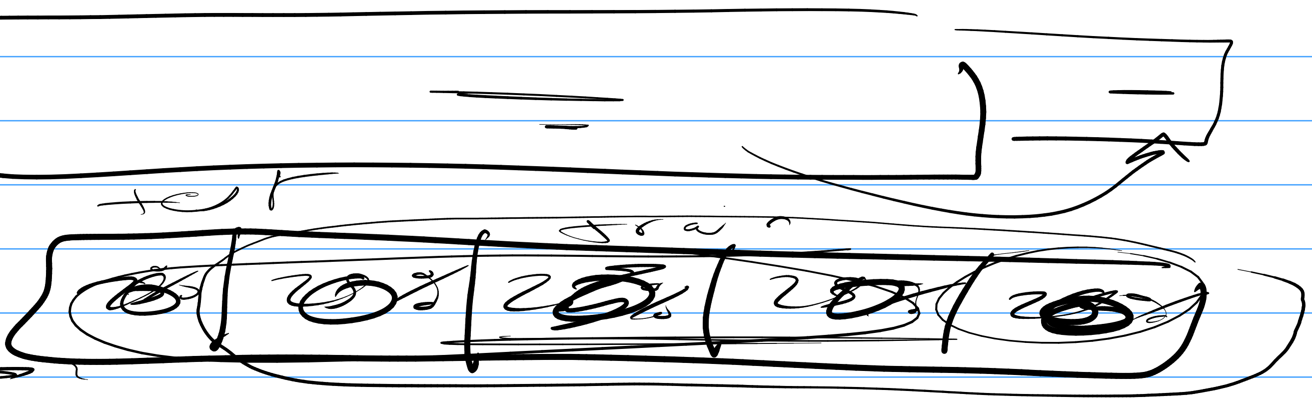
Stratified K Folds

100 50 +ve 50 -ve

80 +ve
20 -ve

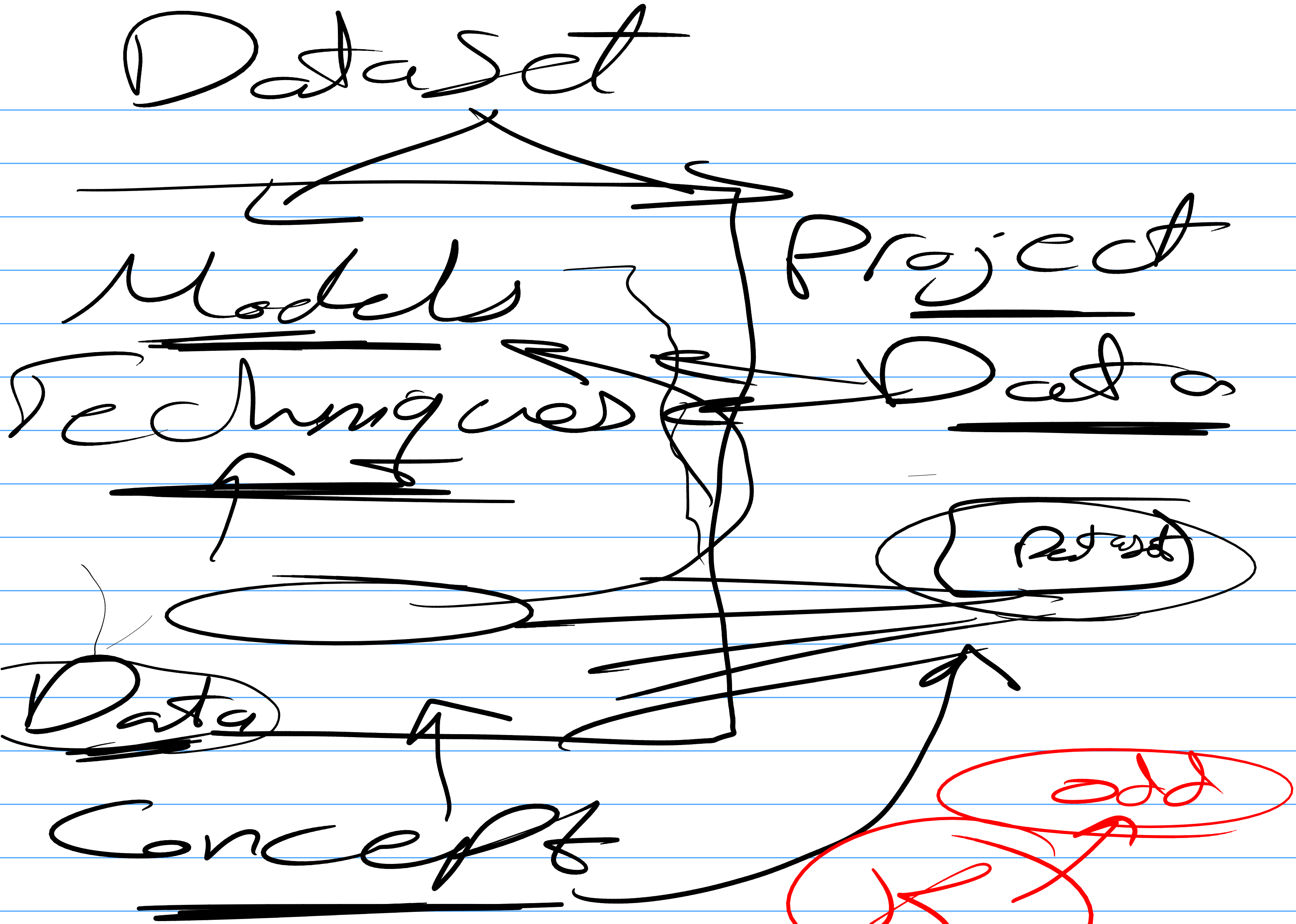
15 +ve
5 -ve

80% +ve 20% -ve



20 +ve

15 +ve 10 +ve
5 -ve 10 -ve
20 +ve



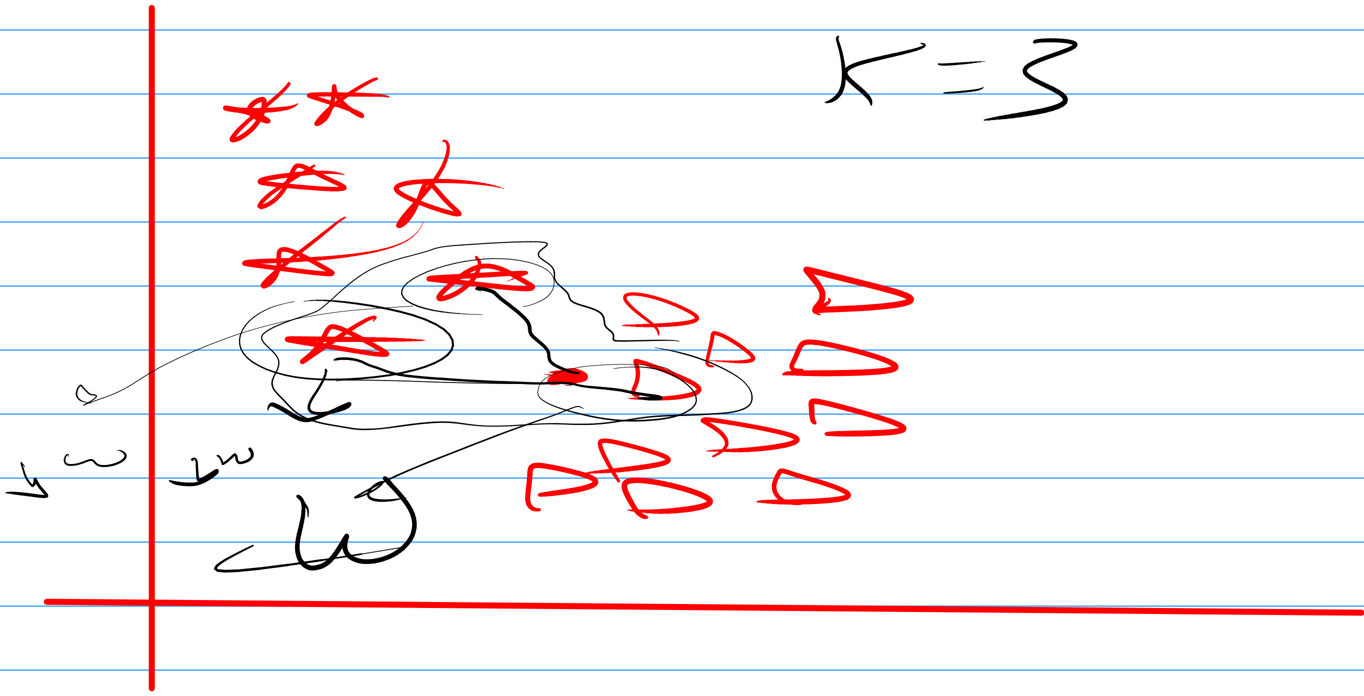
weighted K-NN

K = 4

S 22 21 1 2

$$\downarrow w = \frac{1}{distance \uparrow}$$

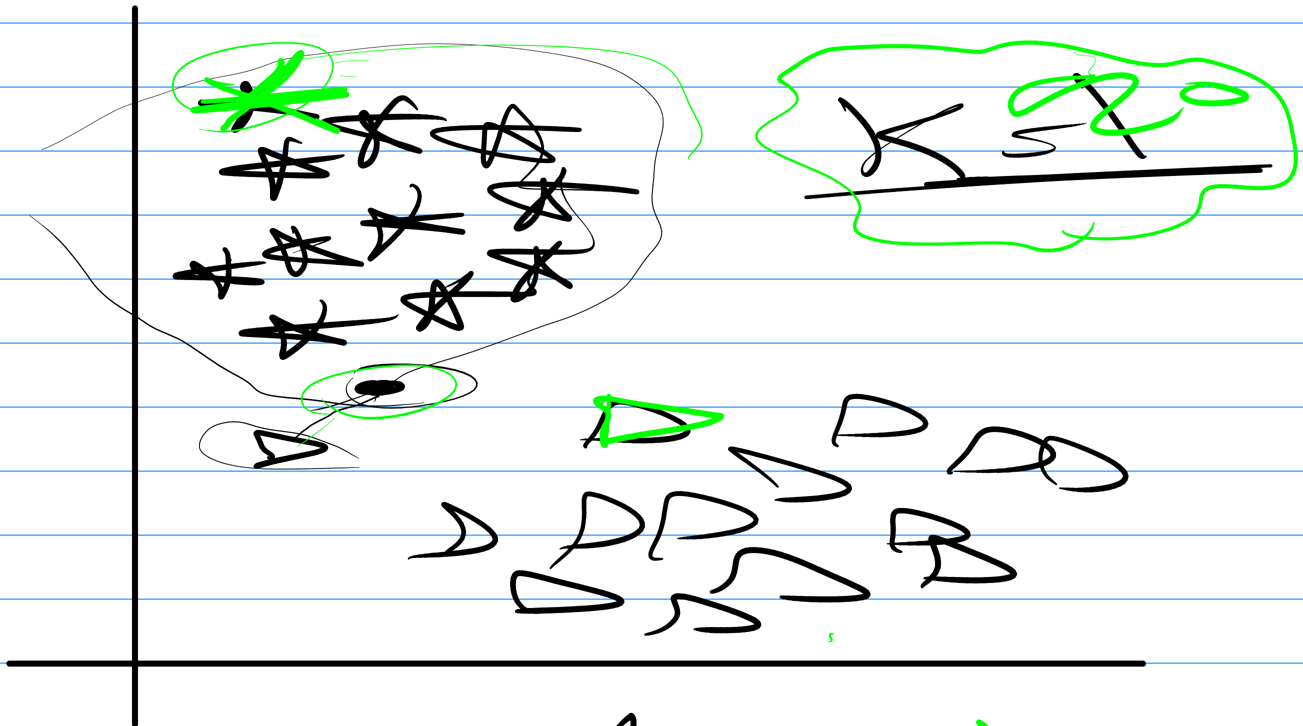
$$\uparrow w \quad \frac{1}{dis \downarrow}$$



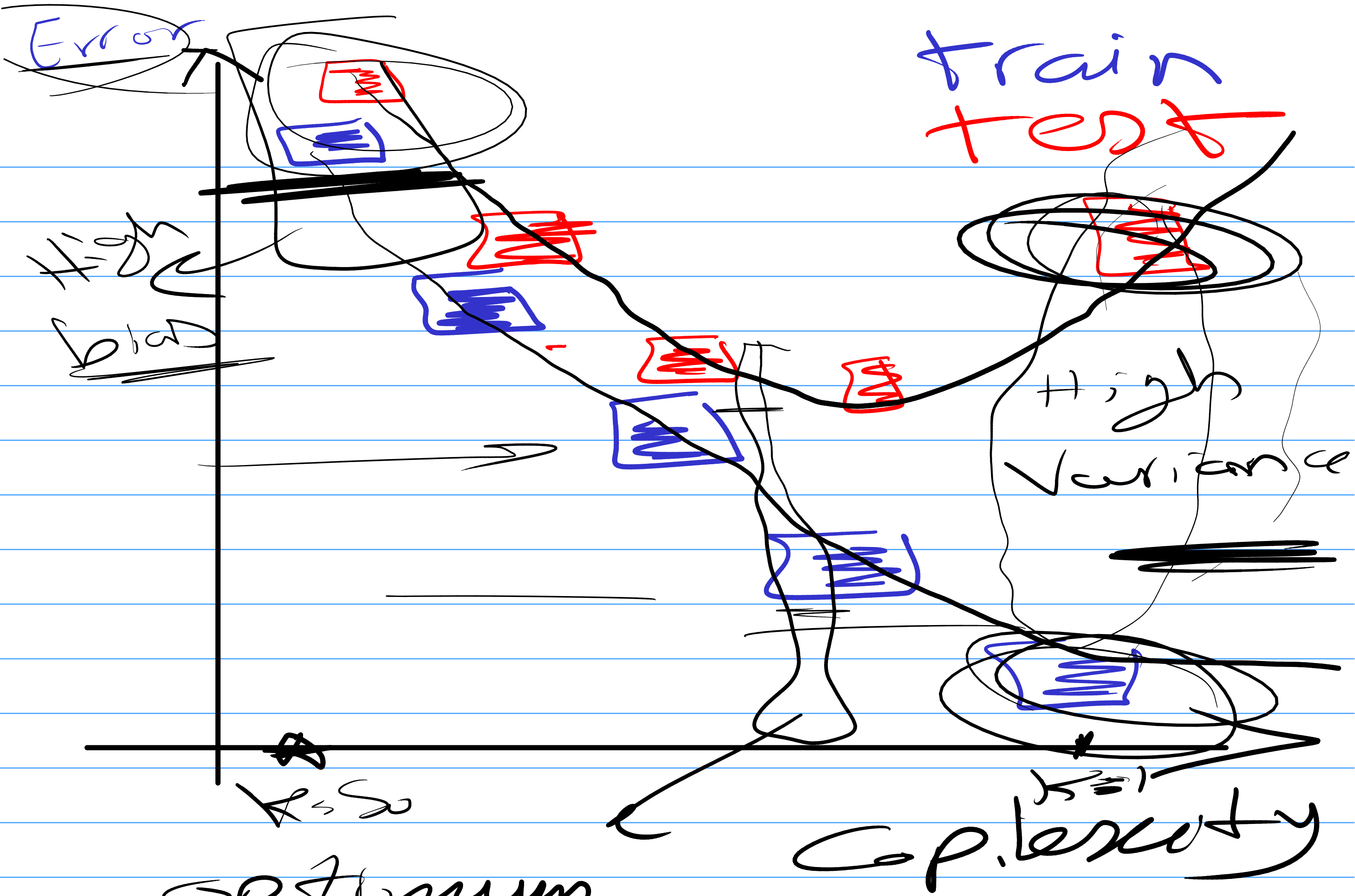
K → overfit
→ Underfit

overfit
↳ train ↑
↳ test ↓

Underfit
↳ train ↓
↳ test ↓



test ↓ train ↑



Optimum Value (Range) of

K-NN model

Prop. Naive Bayes Model

Bayes Theorem

X_1	X_2	X_3	X_4	Y
				0
				1
				0
				0
				1

$$P(y=0 \mid X) = 90\%$$

$$P(y \mid X) = 10\%$$

$$p(y=0 \mid x=(0,2))$$

$$0 - 1 = \frac{p(x=(0,2) \mid y=0)\, p(y=0)}{S\, p(\cancel{x=(0,2)})}$$

$$p(y=1 \mid x=(0,2))$$

$$0 - 1 = \frac{p(x=(0,2) \mid y=1)\, p(y=1)}{S\, p(\cancel{x=(0,2)})}$$

$$p(y=0 \mid x=(0,2))$$
$$= p(x=(0,2) \mid y=0)\, p(y=0)$$

$$p(y=1 \mid x=(0,2))$$
$$= p(x=(0,2) \mid y=1)\, p(y=1)$$

AWS, GCP, ~~AB~~

Cloud Microsoft

as a Service

Bayes

$$P(y \mid x) = \frac{P(x \mid y)\, P(y)}{P(x)}$$

$P($ $\quad$ 2

$x \sim (0, 2)$

$(0, 3)$

$$P = \frac{1}{\infty} = \text{zero}$$

$$\frac{1000000}{\infty} = \text{zero}$$

Descret. Zeit. in

$$\frac{20}{60}$$

20 30 31 40

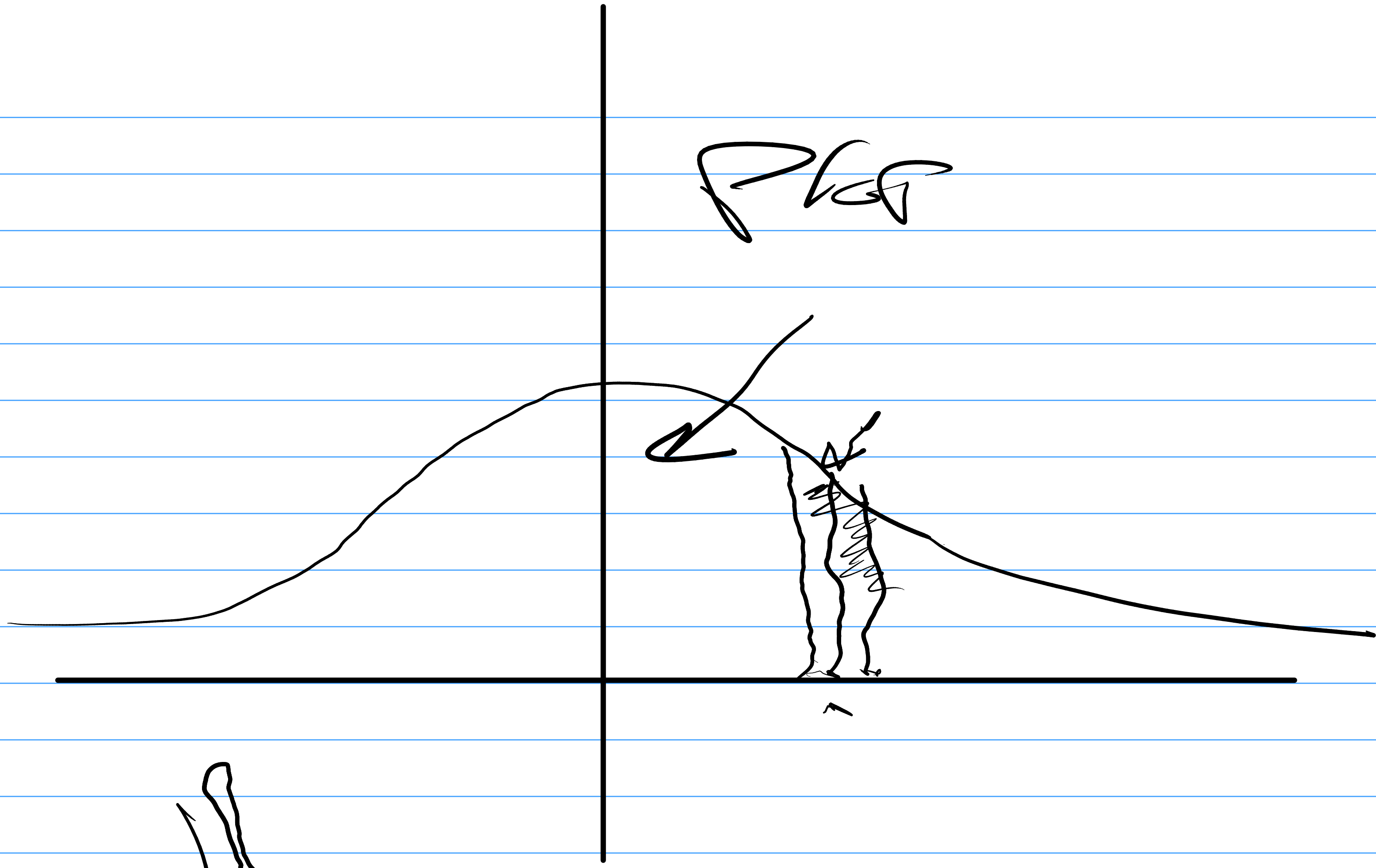
1 2

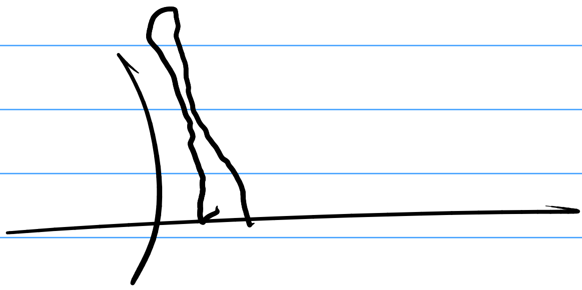
41 50 51 60

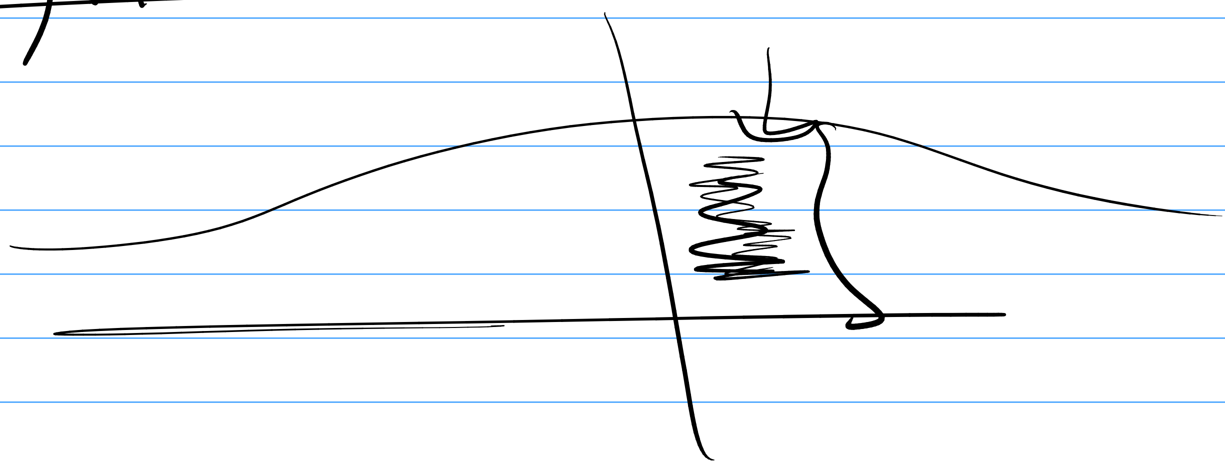
3 4

$$\frac{1}{4}$$

$$\frac{\cancel{1}}{60} \Rightarrow$$ Ze







Cont.

→ GNB

PDF

KNN

x_2 x?

unif

PDF

x → Model

KNN

ML →

I like NLP
and Computer
Vision also
I like ML

2 2 1 1 1 1

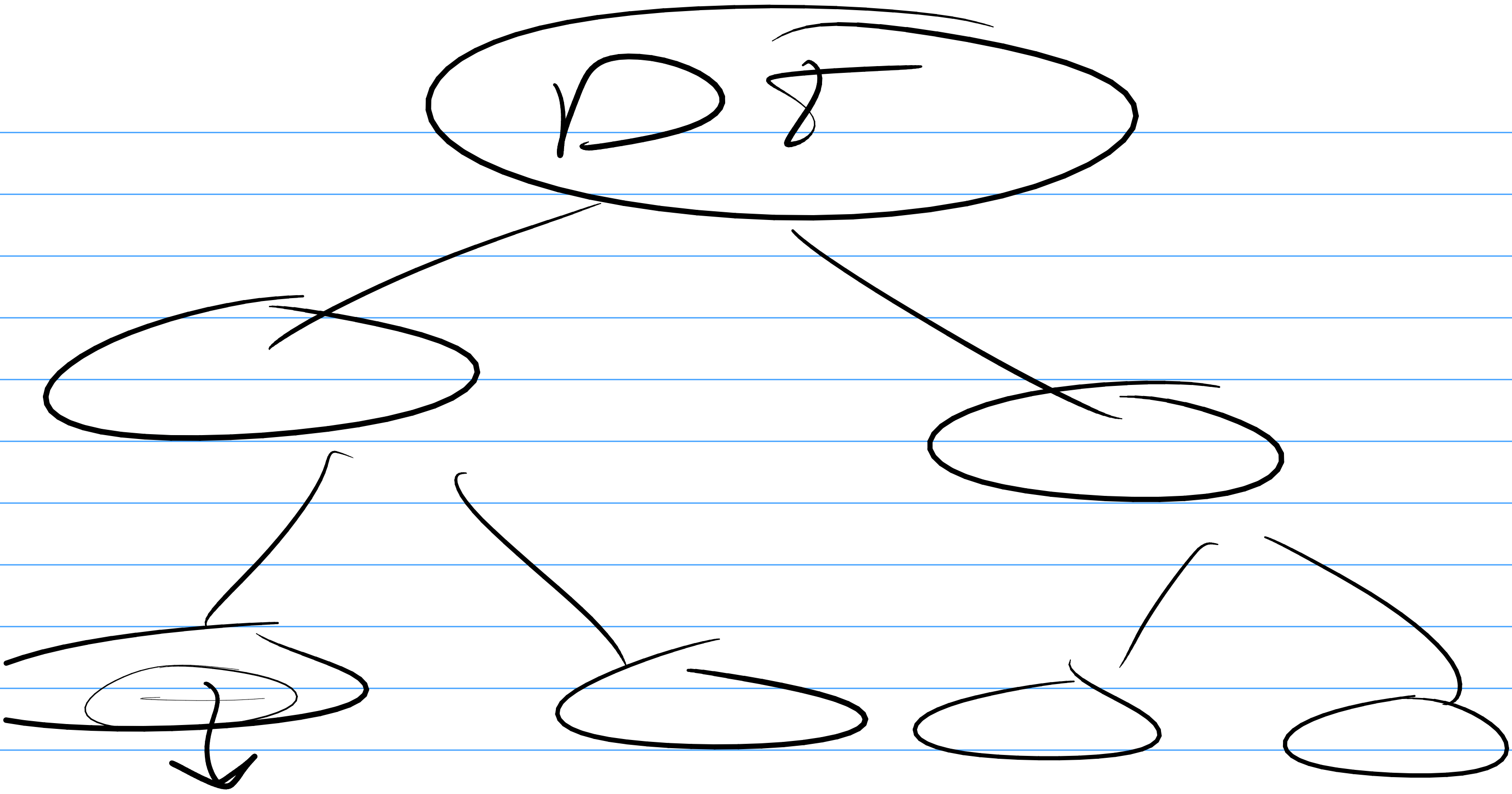
2

mean
↑
K-NN Regr

Naive Bayes →

DT

mean

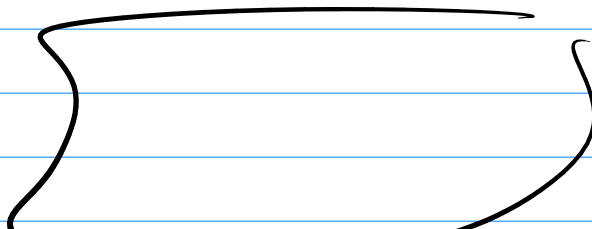


Mean

y → Cont

Concept

Page

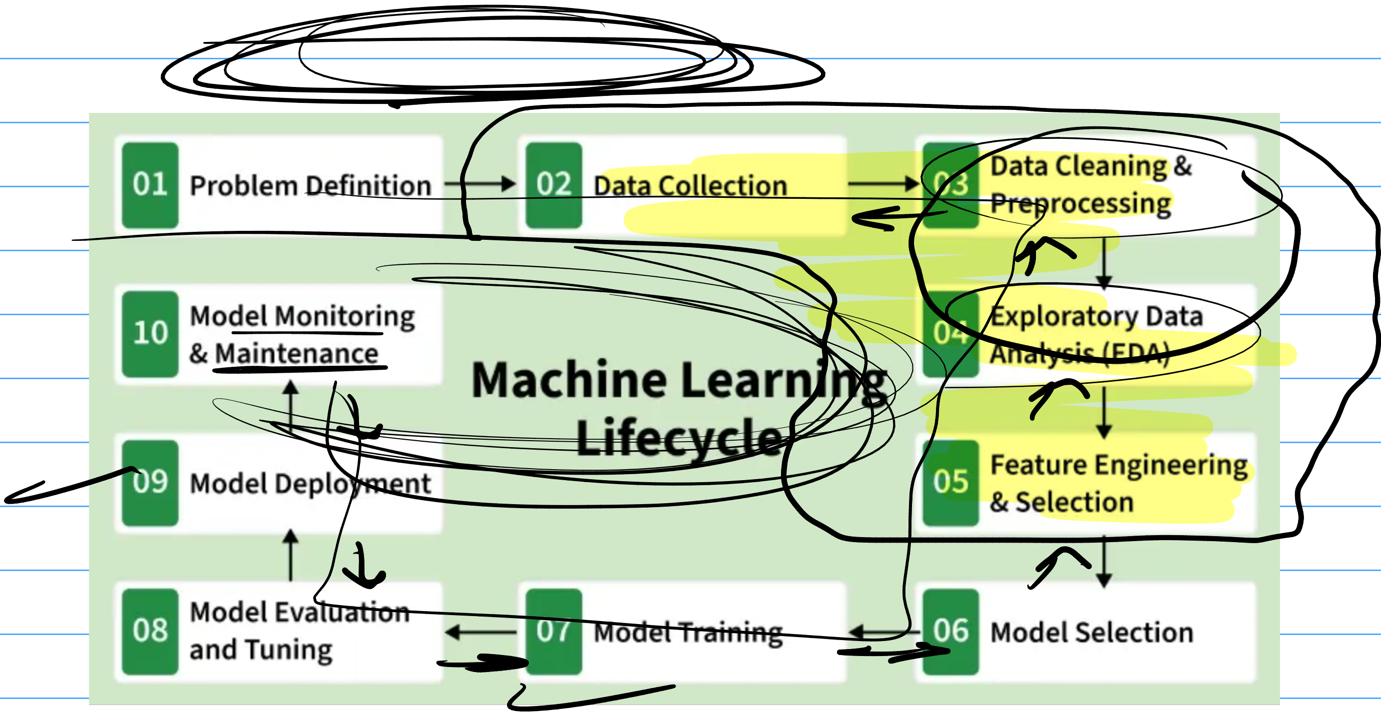


Pipeline

Web Scraping

HTML

Clean



Images — NN ← min max

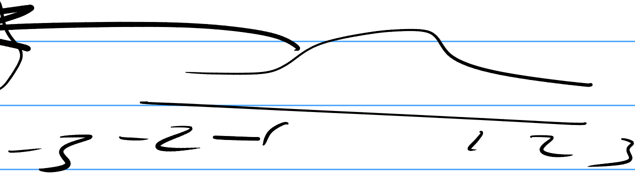
X_{train}

0 — 1

$\mu = 0$

$\sigma = 1$

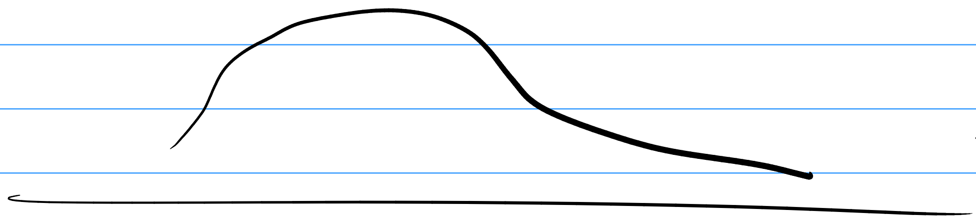
normally distributed ←

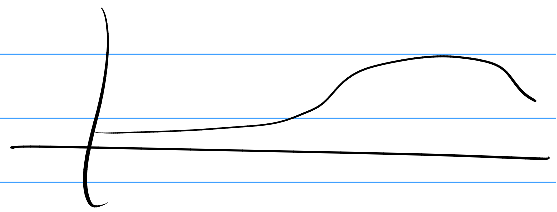


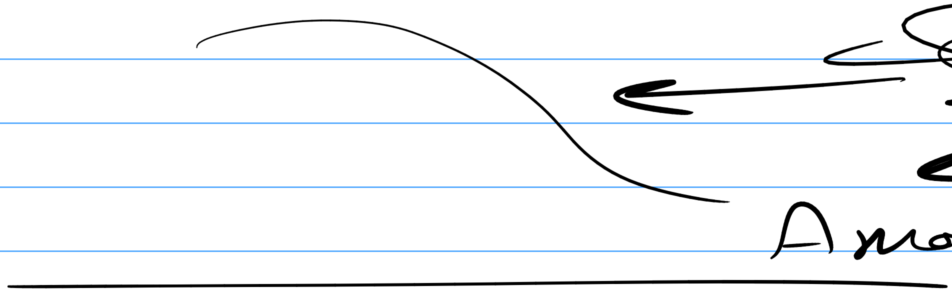
→ normally dist

Scaling

Ages	Salary
20	20000
15	30000
30	40000
40	50000
	60000







Outlier

Anomaly

$X_{train} = 1, 2, 3, 4, 5, 6$

0.2

$X_{test} = 2, 3, 5$

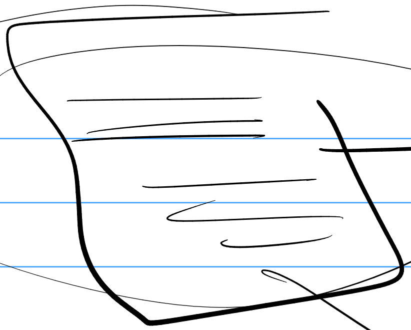
0.2

$$\frac{2-1}{6-1} = \frac{1}{5}$$

$$\frac{X - X_{min}}{X_{max} - X_{min}}$$

Pipeline → steps

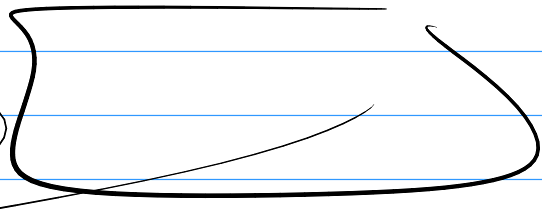
Deployment



Production

deployment

ML



deployment

testing